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Meaningful comparisons between individuals or groups of individuals depend on the assumption that psychological constructs are measured in the same way across various contexts. Unlike weight or height of a person, constructs such as intelligence, attitudes, or personality traits are not directly observable. Instead, researchers infer them from patterns of responses to observed indicators using measurement models. Ensuring that these measurements operate equivalently is therefore a prerequisite for valid scientific conclusions.

Recent methodological discussion have highlighted the need for better statistical methods (Flake et al. 2017; Haig 2020; Altoè et al. 2020) as well as more explicit theoretical foundations in psychology (Eronen and Bringmann 2021). Although these discussions often focus on issues of statistical inference or theory construction, both ultimately depend on the quality of measurement.

As a consequence, neither statistical sophistication nor a strong theoretical framework can compensate for inadequate measurement. When a measure fails to represent the intended construct or functions differently across individuals, groups, or contexts, even highly rigorous analyses may produce misleading conclusions (Flake and Fried 2020; Flake et al. 2017). Thus, a pivotal aspect of measurement quality is *measurement invariance* (MI), which refers to the assumption that a measurement model operates equivalently across individuals, groups, contexts, or measurement occasions.

When MI holds, the relationship between a latent construct and its observed indicators remains stable across the conditions being compared (Meredith 1993). The MI assumption is essential whenever researchers seek to compare latent means, variances, covariances, or structural relations, making MI a fundamental prerequisite for correlational, experimental, and longitudinal research designs (Putnick and Bornstein 2016; Little 1997).

Violations of MI imply that the relationship between the latent construct and its observed indicators differs across groups or conditions. In such cases, differences in observed responses may not solely reflect differences in the underlying construct but may also arise from systematic differences in item functioning or measurement precision. Consequently, observed group differences become ambiguous, as they may represent a combination of substantive construct differences and measurement-related distortions (Cheung and Rensvold 2002; Meredith 1993). Without establishing MI, it is therefore impossible to determine whether observed differences reflect true differences in the latent construct or artifacts of the measurement process. Consequently, MI is not only a methodological requirement but also a prerequisite for valid theoretical interpretation.

Classical Measurement Invariance Testing

Structural equation models (SEMs) are widely used to define latent constructs and their relations to observed indicators (Kline 2023; Bollen and Noble 2011; Tarka 2018). Within SEMs, measurement models specify the relationships between latent factors and their manifest indicators. By separating construct-related variance from measurement error, latent variable models provide a principled basis for evaluating measurement quality (Bollen 1989).

Classical approaches to measurement invariance testing are fundamentally confirmatory in nature. Researchers begin by specifying a measurement model, typically in the form of a confirmatory factor analysis (CFA), that reflects substantive hypotheses about the construct under investigation. A categorical grouping variable is then used to partition the sample into two or more groups, across which measurement parameters are constrained and compared. This most widely used approach for evaluating measurement invariance is usually referred to as multiple-group confirmatory factor analysis (MGCFA) (e.g. Cheung and Rensvold 2002; Putnick and Bornstein 2016).

In MGCFA, MI is typically assessed through a sequence of nested model comparisons in which increasingly restrictive equality constraints are imposed on the model parameters. Equality of the models structure in both groups is called *configural invariance*, and can be understood as both groups sharing the same factor structure (i.e., same number of latent factors and same pattern of loadings between factors and items). Equality of factor loadings former is referred to as *metric invariance*, if metric invariance is present it indicates that all factor loading parameters are equal across groups. Specifically, this suggests that the items of a given scale and their underlying constructs relate to each other with the same strengths in both groups. *Scalar invariance* further constrains item intercepts to equality across groups. Establishing scalar invariance is required for meaningful comparisons of latent means because it indicates that the measurement scale has the same unit and origin across groups (Little 1997). *Residual* or *strict invariance* assumes, additionally to the constraints above, that variables residuals are equal across the groups. If residual variance holds, the items of the scale are measuring the latent constructs with the same measurement error. This implies that it can indeed be known whether differences measured by the items of the tested scale between the groups are due to group differences in the proposed factors or due to measurement errors (Cheung and Rensvold 2002; Widaman and Reise 1997). The appropriate level of invariance required depends on the substantive inference of interest. For example, scalar invariance is generally required when comparing latent means across groups, whereas less restrictive forms of invariance may be sufficient for research questions about correlations (Putnick and Bornstein 2016).

In MGCFA, measurement invariance testing depends on the specified measurement model and grouping variable. Accordingly, measurement noninvariance (MNI) can only be detected with respect to the grouping variables included in the analysis. This framework is well suited to settings in which theory implies meaningful discrete distinctions, such as treatment conditions, demographic categories, or measurement occasions. However, many characteristics that may affect measurement functioning vary continuously across individuals. Converting such characteristics into predefined groups may require arbitrary cutoffs and obscure relevant variation in measurement parameters.

Those limitations of group-based approaches to MI reflect a broader issue concerning the representation of individual differences in psychological research. Developmental and lifespan perspectives have long emphasized that psychological processes often vary continuously across individuals rather than occurring in discrete categories. In particular, lifespan theories highlight interindividual differences in intraindividual change and conceptualize heterogeneity as a fundamental characteristic of human development (**Baltes1987?**; Baltes et al. 2007).

In line with the argument from (**Nesselroade?**), that understanding psychological processes requires statistical models capable of representing systematic individual differences, heterogeneity is an important source of substantive information in its own right (Nesselroade 1991, 2001).

Moderated Nonlinear Factor Analysis

Related arguments have also been made within latent-variable modeling frameworks.

Consistent with lifespan-developmental perspectives emphasizing individual heterogeneity, Little (1997) argued that substantive psychological questions frequently concern individual differences in developmental and psychological processes and therefore require statistical models capable of representing systematic heterogeneity among individuals. Rather than assuming homogeneous relations between latent constructs and observed indicators within broad groups, this perspective encourages the examination of how these relations may vary as a function of individual characteristics (Little 1997). Viewed from this perspective, MNI can be understood as a manifestation of systematic individual differences in the measurement process itself. Addressing such heterogeneity requires models in which measurement parameters can vary directly with individual characteristics rather than only across predefined groups. Moderated nonlinear factor analysis (MNLFA) meets this requirement by modeling factor loadings, intercepts, and other measurement parameters as functions of observed covariates, including continuous characteristics (Bauer and Hussong

2009; Bauer 2017). In this way, MNLFA extends conventional group-based invariance testing by representing measurement heterogeneity without imposing arbitrary group boundaries. Then, MNI is conceptualized as moderation of the measurement model itself. If measurement parameters differ as a function of the moderator, the measurement model no longer operates equivalently across individuals and therefore violates MI (Bauer 2017; Kolbe et al. 2024).

Whether MI is present depends on which parameters are moderated by exogenous variables. The exogenous variables may moderate any subset of the parameters in the measurement model and structural model. Specifically, the means, variances, and covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable (Bauer and Hussong 2009; Bauer 2017). If moderation is limited to the latent-factor parameters, that is, the means, variances, and covariance, the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI (Bauer 2017; Kolbe et al. 2024). Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this conceptualization,this method allows researchers to model gradual, nonlinear, and individual-specific forms of MNI that are difficult to represent within traditional multiple-group approaches (Bauer and Hussong 2009; Kolbe et al. 2024).

This increased flexibility comes at the cost of greater specification requirements. Like MGCFA, MNLFA constitutes a fundamentally confirmatory approach to the assessment of MI (Bauer 2017). In practice, at least three sets of decisions are required. First, the measurement model must be specified, including the latent structure and the relationships between latent and observed variables. Second, one or more moderator variables must be selected to represent potential sources of heterogeneity in the measurement process. Third, moderation functions must be specified for all parameters that are allowed to vary as a function of the moderator. This includes determining which parameters are moderated as well as specifying the functional form of each moderation effect (e.g., linear, quadratic, or another parametric form). Consequently, the detection of MNI depends not only on the choice of moderator variables but also on the assumptions made about how these moderators influence the model parameters (Bauer and Hussong 2009; Bauer 2017). Taken together, MLNFA is a parametric framework for investigating continuous moderators of the measurement model. But what could be done to explore types of moderations when we can neither make assumptions about which of the available covariates moderate the measurement model parameters nor what functional form may be appropriate?

Structural Equation Model Trees

Structural Equation Model trees (SEM trees) were originally introduced by Brandmaier et al. (2013) as a method for identifying important covariates and their interactions within SEMs to explain heterogeneity, e.g., by identifying moderators of model parameters in a SEM. Subsequent developments further refined the methodology and expanded its applicability (Arnold et al., 2021; Brandmaier et al., 2016). Notably, although MI was used as an illustrative application by Arnold et al. (2021), SEM Trees were not originally developed as a dedicated method for testing MI and have received relatively little attention as tools for detecting MNI. Because violations of MI manifest as systematic heterogeneity in measurement parameters, SEM trees appear conceptually well suited for detecting such effects.

SEM trees merge two paradigms: SEMs and on decision trees. Decision trees describe recursive partitions of the covariate space, whereby the sample is repeatedly split according to covariate values into progressively more homogeneous subsets, in order to maximize differences in a given outcome variable. SEM Trees combine the hierarchical structure of decision trees with SEMs, so that they describe heterogeneity in data with respect to a specified SEM as targeted outcome structure. Unlike many other tree-based methods, SEM

Trees can be applied to models that include latent variables.

In contrast to MGCFA and MNLFA, SEM trees represent a more exploratory approach to the assessment of MI. Similar to these methods, SEM trees require the researcher to specify a measurement model a priori. Beyond this, however, the researcher only needs to provide a set of candidate variables that may account for heterogeneity in the data. These variables may be numerous and can be either categorical or continuous. .

Similar to MGCFA, researchers first specify a measurement model and identify variables that may account for heterogeneity. The crucial difference is that MGCFA requires the researcher to select a specific grouping variable and define its functional relationship as moderator in advance, whereas SEM trees can evaluate a large set of candidate variables simultaneously and determine which, if any, are associated with violations of MI. Moreover, SEM trees are able to identify nonlinear forms of moderation and data-driven partition points for continuous variables without requiring these relationships to be specified a priori.

SEM trees combine a theory-driven measurement model with a data-driven search for heterogeneity. This makes them especially valuable when researchers have a well-established measurement model but little prior knowledge about where measurement noninvariance may arise or what form it may take. At the same time, the approach remains closely aligned with familiar MGCFA practice, making SEM trees a comparatively accessible extension of conventional measurement invariance analysis.

At the same time, SEM trees offer substantially greater flexibility by accommodating multiple candidate moderators, continuous and categorical partitioning variables, and potentially nonlinear forms of heterogeneity – given that the sample size is large enough to detect such effects. Although explicit specification of moderators and moderation functions remains advantageous when strong theoretical expectations are available, SEM trees provide a complementary strategy for situations in which the sources or functional forms of heterogeneity are uncertain.

Present Study

More generally, MGCFA, MNLFA, and SEM trees can be understood as reflecting different assumptions regarding prior knowledge about measurement heterogeneity. MGCFA assumes that relevant sources of heterogeneity can be represented through predefined groups. MNLFA further assumes that researchers can identify relevant moderators and specify how these moderators influence model parameters. SEM trees, in contrast, place fewer assumptions on the structure of heterogeneity and instead emphasize data-driven identification of parameter differences. Consequently, the relative performance of these approaches is likely to depend on the extent to which researchers possess accurate prior knowledge about the sources and functional forms of MNI.

The present simulation study compares the performance of MNLFA and SEM trees in detecting MNI across a range of conditions that vary in sample size, effect magnitude, and the location and functional form of moderation. To reflect both the methodological capabilities of the investigated approaches and plausible empirical scenarios, the simulation incorporates different functional forms of MNI, including linear, quadratic, and sigmoid moderation processes. We expect SEM trees to perform particularly well in conditions characterized by nonlinear moderation, especially sigmoid relationships, whereas their relative advantage should diminish in purely linear settings. In contrast, MNLFA is expected to perform best when the functional form of moderation is correctly specified, particularly in linear conditions, but to be more sensitive to misspecification of the underlying moderation function. For both methods, stronger moderation effects are expected to increase the detectability of measurement non-invariance.

To enhance the study's relevance for empirical practice, the simulation is grounded in a realistic applied-research perspective. Specifically, we consider researchers whose experience with MI testing is rooted in the traditional MGCFA framework and who seek to adopt more flexible approaches for detecting heterogeneity. Accordingly, MNLFA and SEM trees are evaluated not only as statistical procedures but also as methodological alternatives for researchers moving beyond conventional MGCFA-based invariance testing.

That is, there are several alternative methods that share some of the advantages highlighted above, a few of which we also want to mention.

Related tree-based approaches have recently been developed, building directly on the SEM-tree framework, specifically for investigating MNI across multiple covariates. Sterner and Goretzko (2023) introduced exploratory factor analysis (EFA) trees as an extension of SEM trees, and Goretzko et al. (2026) subsequently extended this approach to CFA trees. Like SEM trees, EFA and CFA trees use model-based recursive partitioning and score-based structural-change tests to identify parameter instability across candidate covariates. Their principal extension concerns the measurement model to which recursive partitioning is applied: EFA trees integrate an exploratory factor model, facilitating the investigation of MNI during early stages of measurement development, whereas CFA trees incorporate a constrained factor model and thereby allow MNI to be investigated when a hypothesized measurement structure is already available (Sterner and Goretzko (2023); Goretzko et al. (2026)). Thus, SEM trees constitute the more general framework from which these approaches developed and can be applied to parameter heterogeneity in structural equation models beyond factor-analytic models and MI (Brandmaier et al. (2013); Arnold et al. (2021)). In the present study, we apply this general SEM-tree framework specifically to MI by focusing score-based instability tests on factor loadings and item intercepts, corresponding to metric and scalar MNI, respectively. Thus, rather than proposing a dedicated tree-based MI procedure, we investigate the performance of the general SEM-tree framework for detecting MNI and compare it with the parametric moderation approach of MNLFA.

It is to be noted that regularization provides an additional extension to several of these approaches and may be particularly useful when complex measurement models or large numbers of potential sources of MNI are considered. Within EFA, regularization can be used to obtain sparse and interpretable loading structures without relying on conventional factor rotation (Goretzko (2025)), whereas within MNLFA, regularization has been applied to moderation parameters to identify measurement bias across multiple categorical and continuous background characteristics (Bauer et al. (2020); Belzak and Bauer (2024)). Although such extensions can reduce the need to prespecify individual sources of measurement heterogeneity, they also introduce additional methodological choices concerning penalization and model selection. The present study therefore focuses on MNLFA and SEM trees without regularization. This choice reflects our aim to compare approaches that constitute relatively direct extensions of conventional MGCFA for applied researchers with an established measurement model, while differing in the assumptions they require about potential moderators and the functional form of MNI.

Application Example

We want to show the different approaches performance on an illustrative empirical example. For that we chose to explore the MI of the CASP-12 scale regarding age, gender and cultural background as well as additional covariates from the ninth wave of the Survey of Health, Ageing and Retirement in Europe study (SHARE) (SHARE-ERIC 2024). The CASP scale was originally designed to evaluate older peoples quality of life. The CASP-12 is a shorter version of the original 19 item version of the casp scale. Additionally the shorter CASP-12 version, suggested by Wiggins et al. (2008) is also slightly different from the version used in the SHARE assessment (Borrat-Besson et al. 2015; Wiggins et al. 2008).

Bonneville-Roussy et al. (2024) investigated the MI of the CASP-12 scale and its four-factor structure (i.e. Control, Autonomy, Self-Realization, Pleasure), used in the SHARE data with respect to longitudinal invariance, culture, age and gender.

Age

Bonneville-Roussy et al. (2024) looked at the CASP-12's age invariance over time across four age groups: under 60, 60–69, 70–79, 80 and older.

Gender

Gender is assessed by SHARE at the first time of measurement with a coding of 0 for males and 1 for females.

Culture

Cultural invariance was tested between “Western”, “Northern”, “Southern” and “Eastern Europe”.

This highlights the implicit need for categorization, when using MGCFA to test for MI. We want to use this as a possible example case to use these methods for MI testing, that allow for the variables to be included as continuous variables, possibly avoiding arbitrary categorization. Bonneville-Roussy et al. (2024) investigated the longitudinal as well as the cross-sectional MI. For this empirical example we will investigate only the cross-sectional MI of the CASP-12 scale in the ninth wave, with age, gender and culture as covariates.

Method

Simulation

Data-generating mechanism

Data were generated from a single-factor CFA model measured by four continuous indicators. Baseline factor loadings were fixed to

$$\lambda = 0.70,$$

baseline intercepts were fixed to one, and the latent factor variance was fixed to one.

Residual variances were fixed such that indicator reliability equaled $\rho = .75$. at the baseline factor loading of $\lambda = .70$. Because residual variances were held constant while loadings varied under loading moderation, conditional indicator reliability varied accordingly in conditions with metric noninvariance. MNI was introduced by allowing selected factor loadings and/or item intercepts to depend on the moderator.

Nine population models were investigated.

Model	Loading moderation	Intercept moderation
0	none	none
1.1	all items	none
1.11	none	all items
1.12	all items	all items
1.2	items 1–2	none
1.21	none	items 1–2
1.22	items 1–2	items 1–2
1.3	all items; interaction term	none
1.32	loading moderation by one moderator; intercept moderation by a second moderator	

MNI was generated by allowing selected measurement parameters to vary continuously as deterministic functions of an observed moderator variable. For each simulated individual, a moderator

$$M \sim \mathcal{U}(-1, 1)$$

was sampled from a continuous uniform distribution and transformed into an effective moderator

$$Z = h(M),$$

where the transformation $h(\cdot)$ determined the functional form of moderation. Measurement parameters were subsequently modeled as linear functions of the transformed moderator. Consequently, any nonlinearity in parameter variation arose exclusively through the transformation $h(\cdot)$.

Functional forms of moderation

Four moderator functions were considered:

1. **Linear:**

$$h(M) = M$$

In this case, the effective moderator is bounded to the interval $(-1, 1)$ and rotation symmetric around zero.

2. **Sigmoid:**

$$h(M) = a + \frac{b-a}{1+\exp\Big(-k(M-c)\Big)}, \qquad a = -1,\ b = 1,\ c = 0,\ k > 0.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator, rotation symmetric around zero, exhibiting diminishing sensitivity at extreme values.

3. **Quadratic:**

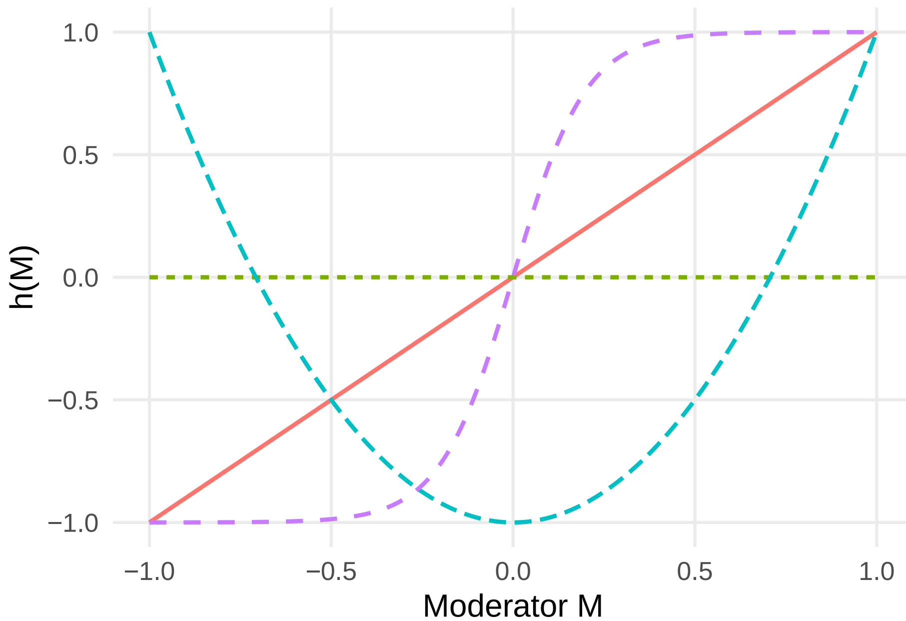
$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. **Noise:**

$$h(M) = 0$$

These functions were selected to represent qualitatively different forms of heterogeneity. The first three functions represented structured forms of moderation, whereas the noise condition represented the absence of systematic moderation. All moderator transformations were bounded to the interval $(-1, 1)$, implying identical maximum parameter deviations for a given moderation coefficient, but not necessarily the same variance. Consequently, identical values of the moderation coefficient imply identical maximum parameter deviations but not necessarily identical average parameter variation.



Simulation design

The simulation crossed data-generating conditions with analysis method. Data-generating conditions manipulated

- population model,
- functional form of moderation (linear, quadratic, sigmoid, null),
- loading moderation magnitude

$$\delta_\lambda \in \{0.20, 0.30\},$$

- intercept moderation magnitude

$$\delta_\nu \in \{0.50, 1.00\},$$

- sample size

$$N \in \{300, 500, 700, 1000\}.$$

Each generated dataset was subsequently analyzed using three methods: linear MNLFA, quadratic MNLFA, and SEM Trees. The resulting factorial design consisted of 576 factorial design cells, each of which was analyzed using three analysis methods.

Analysis models

Each simulated dataset was analyzed using both MNLFA (i.e. either 'linear MNLFA' or 'quadratic MNLFA') and score-based SEM Trees.

Two MNLFA specifications were considered. In the linear MNLFA analysis, moderation was modeled as a linear function of the transformed moderator, which we consider the default use case of MLNFA. In the quadratic MNLFA analysis, moderation was modeled using the quadratically transformed moderator, assuming that a researcher expected a quadratic effect of the moderator. Thus, the analytical model was correctly specified whenever the assumed moderator transformation matched the data-generating process and misspecified otherwise.

For SEM trees, the same basic measurement model was estimated using recursive partitioning. To ensure a fair comparison with MNLFA, SEM trees were supplied only with the true moderator variables. Consequently, differences between both methods reflect differences in how moderation is modeled/handled rather than differences in the available information.

In an additional robustness analysis, the true moderator was supplemented with ten independent noise predictors. This condition was included to investigate the ability of SEM Trees to distinguish informative moderators from irrelevant covariates when multiple candidate splitting variables are available.

Evaluation criteria

Performance was evaluated separately for metric and scalar MNI. For MNLFA, metric and scalar MNI were assessed using sequential likelihood-ratio tests comparing nested moderation models. For SEM trees, MNI was assessed using score-based global parameter-instability tests on the respective focus parameters (factor loadings or intercepts) with Bonferroni-adjusted significance tests. [TODO: AB->LH: Zitationen zu den score-based Tests, Arnold und Zeileis einfügen] For each population model and simulation condition, the following performance measures were computed:

- **Statistical power:** The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level when the respective form of MNI was present in the data-generating model.

- **Rejection rate:** The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level when that form of MNI was absent in the data-generating model.

In the primary comparison, SEM trees were supplied only with the true moderator variables. Consequently, statistical power and rejection rates reflect the ability to detect MNI independent of moderator selection.

In an additional robustness analysis, the true moderator was supplemented with ten independent noise predictors. For this condition, statistical power was defined as

$$P(\text{reject parameter stability} \cap \text{select true moderator} \mid \text{the corresponding form of MNI is present}),$$

such that a simulation replication counted as successful only if the tree both rejected the null hypothesis of parameter stability and selected the true moderator as a splitting variable. This criterion therefore jointly evaluates the ability of SEM trees to detect MNI and to distinguish informative moderators from irrelevant candidate predictors.

Replications

Each data-generating design cell was replicated 100 times. Every generated dataset was analyzed using all three analysis methods, resulting in 1,728 method-specific analysis cells. An additional 100 SEM tree analyses were conducted for the noisy-predictor sensitivity condition.

Empirical Example

The CASP-12 scale [@] comprises 12 items and assesses four domains of quality of life: Control, Autonomy, Self-Realization and Pleasure. Each factor is assessed using three items. Responses are rated on a four-point rating scale ranging from 1 (Often) to 4 (Never), with some items requiring inverse scoring before calculation of the total score. Higher CASP-12 scores indicate better quality of life. Age as covariate is included continuously, gender is included as dichotomous variable and culture is All covariates are included simultaneously in the analytical model.

Software

All simulations were conducted in R (R Core Team 2025). MNLFA models were estimated using `OpenMx` (Boker et al. 2026) with raw-data maximum likelihood and definition variables. Score-based SEM trees were estimated using the `semtree` package (Arnold et al. 2021). [TODO: AB->LH: Es gibt eine eigene Package citation!] Simulation design construction, data handling, aggregation, and visualization will rely on `dplyr`(Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `future` (Bengtsson 2021), `parallel` (R Core Team 2025), and `ggplot2` (Wickham 2016).

Results

All conditions comprised 100 attempted replications. SEM-tree tests were available for all replications, whereas a small number of MNLFA/MNLFAQ replications did not yield evaluable likelihood-ratio tests. Power and rejection rates were therefore calculated among evaluable replications, with estimation failures reported separately.

```
## Warning: Paket 'knitr' wurde unter R Version 4.5.2 erstellt
```

Population_Model	Truth	Method	Metric	Scalar
0	No MNI	MNLFA	0.063 (0.003)	0.052 (0.003)
0	No MNI	MNLFAQ	0.067 (0.004)	0.053 (0.003)
0	No MNI	SEMTREE	0.050 (0.003)	0.044 (0.003)
1.1	Metric only	MNLFA	0.678 (0.003)	0.068 (0.004)
1.1	Metric only	MNLFAQ	0.393 (0.004)	0.067 (0.004)
1.1	Metric only	SEMTREE	0.930 (0.003)	0.053 (0.003)
1.11	Scalar only	MNLFA	0.067 (0.004)	0.690 (0.002)
1.11	Scalar only	MNLFAQ	0.285 (0.005)	0.372 (0.003)
1.11	Scalar only	SEMTREE	0.247 (0.005)	0.993 (0.001)
1.12	Metric + scalar	MNLFA	0.678 (0.003)	0.693 (0.002)
1.12	Metric + scalar	MNLFAQ	0.639 (0.005)	0.373 (0.003)
1.12	Metric + scalar	SEMTREE	0.916 (0.003)	0.997 (0.001)
1.2	Metric only	MNLFA	0.698 (0.003)	0.063 (0.003)
1.2	Metric only	MNLFAQ	0.418 (0.004)	0.058 (0.003)
1.2	Metric only	SEMTREE	0.976 (0.002)	0.059 (0.003)
1.21	Scalar only	MNLFA	0.204 (0.004)	0.701 (0.002)
1.21	Scalar only	MNLFAQ	0.427 (0.005)	0.375 (0.003)
1.21	Scalar only	SEMTREE	0.477 (0.005)	1.000 (0.000)
1.22	Metric + scalar	MNLFA	0.810 (0.004)	0.702 (0.003)
1.22	Metric + scalar	MNLFAQ	0.798 (0.004)	0.375 (0.003)
1.22	Metric + scalar	SEMTREE	0.968 (0.002)	1.000 (0.000)
1.3	Metric only	MNLFA	0.721 (0.003)	0.080 (0.004)
1.3	Metric only	MNLFAQ	0.441 (0.004)	0.072 (0.004)
1.3	Metric only	SEMTREE	0.972 (0.002)	0.054 (0.003)
1.32	Metric + scalar	MNLFA	0.681 (0.003)	0.696 (0.002)
1.32	Metric + scalar	MNLFAQ	0.596 (0.005)	0.374 (0.003)
1.32	Metric + scalar	SEMTREE	0.769 (0.005)	0.996 (0.001)

Table 1

level	estimand	moderator	MNLFA	MNLFAQ	SEMTREE
Metric	Power	linear	0.970 (0.002)	0.407 (0.003)	0.934 (0.002)
Metric	Power	quadratic	0.166 (0.003)	0.978 (0.001)	0.903 (0.003)
Metric	Power	sigmoid	0.997 (0.001)	0.258 (0.004)	0.928 (0.002)
Metric	Rejection u. invariance	linear	0.058 (0.003)	0.453 (0.004)	0.350 (0.004)
Metric	Rejection u. invariance	quadratic	0.215 (0.004)	0.059 (0.003)	0.293 (0.004)
Metric	Rejection u. invariance	sigmoid	0.062 (0.003)	0.267 (0.005)	0.131 (0.004)
Scalar	Power	linear	1.000 (0.000)	0.066 (0.003)	1.000 (0.000)
Scalar	Power	quadratic	0.089 (0.003)	1.000 (0.000)	0.992 (0.001)
Scalar	Power	sigmoid	1.000 (0.000)	0.054 (0.003)	1.000 (0.000)
Scalar	Rejection u. invariance	linear	0.055 (0.003)	0.057 (0.003)	0.051 (0.003)
Scalar	Rejection u. invariance	quadratic	0.080 (0.003)	0.077 (0.003)	0.050 (0.003)
Scalar	Rejection u. invariance	sigmoid	0.062 (0.003)	0.053 (0.003)	0.056 (0.003)

Table 2

Power and rejection rates under invariance by moderator functional form. Monte Carlo standard errors are given in parentheses.

Truth	Method	Metric decision	Scalar stage reached	Scalar decision	Invariance retained	Correct classification
No MNI	MNLFA	0.063	0.937	0.048	0.889	0.889
No MNI	MNLFAQ	0.067	0.933	0.050	0.882	0.882
No MNI	SEMTREE	0.050	0.950	0.041	0.909	0.909
Metric only	MNLFA	0.699	0.301	0.026	0.274	0.699
Metric only	MNLFAQ	0.418	0.583	0.033	0.550	0.418
Metric only	SEMTREE	0.959	0.041	0.002	0.039	0.959
Scalar only	MNLFA	0.136	0.864	0.645	0.219	0.645
Scalar only	MNLFAQ	0.356	0.644	0.332	0.313	0.332
Scalar only	SEMTREE	0.362	0.638	0.635	0.003	0.635
Metric + scalar	MNLFA	0.723	0.277	0.035	0.242	0.723
Metric + scalar	MNLFAQ	0.678	0.322	0.024	0.298	0.678
Metric + scalar	SEMTREE	0.884	0.116	0.115	0.001	0.884

Table 3

Sequential classification outcomes by true measurement-invariance condition and method.

Simulation

Empirical Example

Discussion

- Possible explanation for the high scalar power of SEM Trees relative to MNLFA:
- The SEM Tree directly tests instability in the item intercept parameters (nu_1, . . . , nu_4).
- In the MNLFA scalar model, direct moderator effects on the item intercepts are constrained, but moderator effects on the latent factor mean remain freely estimated. -> Latent-mean moderation may absorb true intercept moderation in MNLFA.
- In several population models (1.11, 1.12, 1.32), the same intercept moderation effect is applied to all four indicators.
- Such a common shift in observed item means can potentially be represented by a

moderator-dependent latent factor mean.

- This is particularly relevant because the baseline loadings are equal (lambda = .70), making a common latent-mean shift capable of producing similar mean shifts across all indicators.

- Consequently, constraining direct intercept moderation to zero may produce relatively little model deterioration, reducing the power of the MNLFA scalar LRT.

- This problem should be weaker for partial intercept noninvariance.

- A single latent-mean moderation effect affects all indicators through their loadings and therefore cannot reproduce this selective pattern as easily.

- SEM Trees approximate moderator effects through partitions and therefore do not require the true effect to follow a prespecified continuous functional form.

- MNLFA linear and quadratic models impose specific functional forms, potentially reducing power under functional-form misspecification.

- The higher scalar power of SEM Trees should therefore not immediately be interpreted as generally superior sensitivity. At least part of the difference may arise because the MNLFA scalar model can represent common indicator-mean shifts through latent-mean moderation, whereas the SEM Tree directly targets intercept instability.

- Distinction to CFA trees and EFA trees!

Practical Recommendations

References

Altoè, Gianmarco, Giulia Bertoldo, Claudio Zandonella Callegher, et al. 2020. "Enhancing Statistical Inference in Psychological Research via Prospective and Retrospective Design Analysis." *Frontiers in Psychology*, advance online publication.

https://doi.org/10.3389/fpsyg.2019.02893.

Arnold, M., M. C. Voelkle, and A. M. Brandmaier. 2021. "Score-Guided Structural Equation Model Trees." *Frontiers in Psychology* 11: 3913. https://doi.org/10.3389/fpsyg.2020.564403.

Baltes, Paul B., Ulman Lindenberger, and Ursula M. Staudinger. 2007. "Life Span Theory in Developmental Psychology." In *Handbook of Child Psychology*, 1st ed., edited by William Damon and Richard M. Lerner. Wiley. https://doi.org/10.1002/9780470147658.chpsy0111.

Bauer, D. J. 2017. "A More General Model for Testing Measurement Invariance and Differential Item Functioning." *Psychological Methods* 22 (3): 507–26. https://doi.org/10.1037/met0000077.

Bauer, D. J., W. C. M. Belzak, and V. T. Cole. 2020. "Simplifying the Assessment of Measurement Invariance over Multiple Background Variables: Using Regularized Moderated Nonlinear Factor Analysis to Detect Differential Item Functioning." *Structural Equation Modeling: A Multidisciplinary Journal* 27 (1): 43–55. https://doi.org/10.1080/10705511.2019.1642754.

Bauer, D. J., and A. M. Hussong. 2009. "Psychometric Approaches for Developing Commensurate Measures Across Independent Studies: Traditional and New Models." *Psychological Methods* 14 (2): 101–25. https://doi.org/10.1037/a0015583.

Belzak, William, and Daniel Bauer. 2024. "Using Regularization to Identify Measurement Bias Across Multiple Background Characteristics: A Penalized

Expectation–Maximization Algorithm." *Journal of Educational and Behavioral Statistics* 49 (February): 976–1012. https://doi.org/10.3102/10769986231226439.

Bengtsson, Henrik. 2021. "A Unifying Framework for Parallel and Distributed Processing in r Using Futures." *The R Journal* 13 (2): 208–27. https://doi.org/10.32614/RJ-2021-048.

Boker, Steven M., Michael C. Neale, Hermine H. Maes, et al. 2026. *OpenMx: Extended Structural Equation Modelling*. https://doi.org/10.32614/CRAN.package.OpenMx.

Bollen, K. A. 1989. *Structural Equations with Latent Variables*. John Wiley & Sons. https://doi.org/10.1002/9781118619179.

Bollen, Kenneth A., and Mark D. Noble. 2011. "Structural Equation Models and the Quantification of Behavior." *Proceedings of the National Academy of Sciences of the United States of America* 108 (Suppl 3): 15639–46. https://doi.org/10.1073/pnas.1010661108.

Bonneville-Roussy, Arielle, Florence Khoriaty, and François Laberge. 2024. "Time, Age, Gender and Cultural Measurement Invariance of the CASP-12, a Measure of Psychological Quality of Life in Adulthood." *Quality of Life Research* 33: 1569–79. https://doi.org/10.1007/s11136-024-03639-7.

Borrat-Besson, Carmen, Valerie-Anne Ryser, and Judite Goncalves. 2015. *An Evaluation of the CASP-12 Scale Used in the Survey of Ageing and Retirement in Europe (SHARE) to Measure Quality of Life Among People Aged 50+*. FORS Working Paper Series Nos. 2015-4. FORS.

Brandmaier, A. M., T. von Oertzen, J. J. McArdle, and U. Lindenberger. 2013. "Structural Equation Model Trees." *Psychological Methods* 18 (1): 71–86. https://doi.org/10.1037/a0030001.

Cheung, Gordon W., and Roger B. Rensvold. 2002. "Evaluating Goodness-of-Fit Indexes for Testing Measurement Invariance." *Structural Equation Modeling: A Multidisciplinary Journal* 9 (2): 233–55. https://doi.org/10.1207/S15328007SEM0902/_5.

Eronen, Markus I., and Laura F. Bringmann. 2021. "The Theory Crisis in Psychology: How to Move Forward." *Perspectives on Psychological Science* 16 (4): 779–88. https://doi.org/10.1177/1745691620970586.

Flake, Jessica K., and Eiko I. Fried. 2020. "Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them." *Advances in Methods and Practices in Psychological Science* 3 (4): 456–65. https://doi.org/10.1177/2515245920952393.

Flake, Jessica K., Jolynn Pek, and Eric Hehman. 2017. "Construct Validation in Social and Personality Research: Current Practice and Recommendations." *Social Psychological and Personality Science* 8 (4): 370–78. https://doi.org/10.1177/1948550617693063.

Goretzko, David. 2025. *Regularized Exploratory Factor Analysis as an Alternative to Factor RotationRegularized Exploratory Factor Analysis as an Alternative to Factor Rotation.* No. 4. Hogrefe Publishing. https://doi.org/10.1027/1015-5759/a000792.

Goretzko, David, Matt C. Howard, and Philipp Sterner. 2026. "Investigating Measurement Invariance for Multiple Covariates in Organizational Research Using Exploratory Factor Analysis and Confirmatory Factor Analysis Trees." *Journal of Applied Psychology* 111 (8):

1037–53. https://doi.org/10.1037/apl0001368.

Haig, Brian D. 2020. "What Can Psychology's Statistics Reformers Learn from the Error-Statistical Perspective?" *Methods in Psychology* 2: 100020. https://doi.org/https://doi.org/10.1016/j.metip.2020.100020.

Kline, Rex B. 2023. "Structural Equation Modeling." In *The Cambridge Handbook of Research Methods and Statistics for the Social and Behavioral Sciences: Volume 1: Building a Program of Research*, edited by Austin Lee Nichols and JohnEditors Edlund. Cambridge Handbooks in Psychology. Cambridge University Press.

Kolbe, Laura, Dylan Molenaar, Suzanne Jak, and Terrence D. Jorgensen. 2024. "Assessing Measurement Invariance with Moderated Nonlinear Factor Analysis Using the R Package OpenMx." *Psychological Methods* 29 (2): 388–406. https://doi.org/10.1037/met0000501.

Little, Todd D. 1997. "Mean and Covariance Structures (MACS) Analyses of Cross-Cultural Data: Practical and Theoretical Issues." *Multivariate Behavioral Research* 32 (1): 53–76. https://doi.org/10.1207/s15327906mbr3201_3.

Meredith, William. 1993. "Measurement Invariance, Factor Analysis and Factorial Invariance." *Psychometrika* 58 (4): 525–43. https://doi.org/10.1007/BF02294825.

Müller, Kirill, and Hadley Wickham. 2026. *Tibble: Simple Data Frames*. https://doi.org/10.32614/CRAN.package.tibble.

Nesselroade, J. R. 1991. "The Warp and the Woof of the Developmental Fabric." In *Visions of Aesthetics, the Environment, & Development: The Legacy of Joachim f. Wohlwill*, edited by R. Downs, L. Liben, and D. S. Palermo. Lawrence Erlbaum Associates.

Nesselroade, J. R. 2001. "Intraindividual Variability in Development Within and Between Individuals." *European Psychologist* 6 (3): 187–93. https://doi.org/10.1027/1016-9040.6.3.187.

Putnick, Diane L., and Marc H. Bornstein. 2016. "Measurement Invariance Conventions and Reporting: The State of the Art and Future Directions for Psychological Research." *Developmental Review* 41: 71–90. https://doi.org/https://doi.org/10.1016/j.dr.2016.06.004.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

SHARE-ERIC. 2024. "Survey of Health, Ageing and Retirement in Europe (SHARE) Wave 9." Version 9.0.0. SHARE-ERIC. https://doi.org/10.6103/SHARE.w9.900.

Sterner, P., and D. Goretzko. 2023. "Exploratory Factor Analysis Trees: Evaluating Measurement Invariance Between Multiple Covariates." *Structural Equation Modeling: A Multidisciplinary Journal* 30 (6): 871–86. https://doi.org/10.1080/10705511.2023.2188573.

Tarka, Piotr. 2018. "An Overview of Structural Equation Modeling: Its Beginnings, Historical Development, Usefulness and Controversies in the Social Sciences." *Quality & Quantity* 52 (1): 313–54. https://doi.org/10.1007/s11135-017-0469-8.

Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New

York. https://ggplot2.tidyverse.org.

Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026. *Dplyr: A Grammar of Data Manipulation*. https://doi.org/10.32614/CRAN.package.dplyr.

Wickham, Hadley, and Lionel Henry. 2026. *Purrr: Functional Programming Tools*. https://doi.org/10.32614/CRAN.package.purrr.

Wickham, Hadley, Davis Vaughan, and Maximilian Girlich. 2025. *Tidyr: Tidy Messy Data*. https://doi.org/10.32614/CRAN.package.tidyr.

Widaman, Keith F., and Steven P. Reise. 1997. "Exploring the Measurement Invariance of Psychological Instruments: Applications in the Substance Use Domain." In *The Science of Prevention: Methodological Advances from Alcohol and Substance Abuse Research*. American Psychological Association. https://doi.org/10.1037/10222-009.

Wiggins, R. D., G. Netuveli, M. Hyde, P. Higgs, and D. Blane. 2008. "The Evaluation of a Self-Enumerated Scale of Quality of Life (CASP-19) in the Context of Research on Ageing: A Combination of Exploratory and Confirmatory Approaches." *Social Indicators Research* 89: 61–77. https://doi.org/10.1007/s11205-007-9220-5.